Multi-Mode Digital Radio (MMDR) – General data

1. General

The speech communication system is part of the Very High Frequency (VHF) Communication systems 1 and 2 (refer to 23-12-00, VHF Communication System – General Data).

The VHF Communication systems 1 and 2 give voice communication between aircraft, or between the aircraft and a ground station. The two systems (VHF 1 and VHF 2) are the same.

The VHF Communication system 1 and 2 have the following components:

- Two Multi-Mode Digital Radio (MMDR) transceivers
- Two VHF antennas

The MMDR transceiver is a VHF NAV/COM transceiver with a self contained Automatic Direction Finder (ADF) receiver.

The description and operation of the navigation part of the MMDR transceiver is given in this section.

The description and operation of the ADF part of the MMDR transceiver is given in 34-55-00, Automatic Direction Finding System (ADF) – General Data.

The KTR Multi-Mode Digital Radio (MMDR) is a blind rack-mounted integrated transceiver consisting of one transmitter and six receivers.

2. Description

A. Multi-Mode Digital Radio (KTR 2280)

Two KTR 2280 Multi-Mode Digital Radio (MMDR) integrated transceivers are installed in the nose avionics bay. The MMDR comprises of one 16W COM transmitter and six receivers. The receivers consist of a VHF Omni Range (VOR), Glide Slope (GS), Localizer (LOC), and three simultaneously operating COM receivers. Primary controls for the MMDR are on the MF controller and the PF controllers, with display of the selected information on the PFD screen.

28 Volt DC power is protected by a 7.5 Amp MMDR 1 (F3) (Left Bus) and a 7.5 Amp MMDR 2 (F4) (Right Bus) circuit breaker located on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

The Multi-Mode Digital Radio is integrated to the following aircraft systems:

1) Modular Avionics Unit (MAU) (refer to 31-43-00, Modular Avionics Unit (MAU) (Mod 6/2049) – General Data).

2) VHF Navigation (LOC/GS/VOR) (refer to 34-52-00, VHF Navigation System – General Data).

3) VHF Communication (refer to 23-12-00, VHF Communication System – General Data).

4) Automatic Direction Finder (ADF) (refer to 34-55-00, Automatic Direction Finding System (ADF) – General Data), if installed.

3. Adjustment/Test

A. Multi-Mode Digital Radio (KTR 2280)

To function test the VHF Communication, VHF Navigation, and the ADF components of the MMDR refer to the following chapters:

1) VHF Navigation (LOC/GS/VOR) (refer to 34-52-00, VHF Navigation System – General Data, Para 3.A.2)).

2) VHF Communication (refer to 23-12-00, VHF Communication System – General Data, Para 3.A.2)).

3) Automatic Direction Finder (ADF) (refer to 34-55-00, Automatic Direction Finding System (ADF) – General Data, Para 3.A.2)).

4. Removal/Installation

WARNING **INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECT OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:**

CHASSIS GROUNDING – ALL RACK MOUNT / REMOTE MOUNT UNITS SHALL BE ELECTRICALLY BONDED TO THE AIRFRAME. ENSURE THAT THE MATING SURFACES BETWEEN THE LRU MOUNTING TRAY (OR LRU MOUNTING FEET IF A TRAY IS NOT USED) PROVIDE A LOW IMPEDANCE ELECTRICAL PATH. ALL MATING SURFACES MUST BE FREE OF ALL PAINT AND OTHER NONCONDUCTIVE ELEMENTS. REFER TO BONDING REQUIREMENTS IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELDS GROUNDS – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELLS/STRAIN RELIEF AND THE MOUNTING TRAY OR LRU CASE AS PER REQUIREMENTS IN PSM 1-64-2W.

A. Removal/Installation Multi-Mode Digital Radio (KTR 2280)

1) General

These procedures are for the removal and installation of the Multi-Mode Digital Radio (KTR 2280). MMDR 1 and MMDR 2 (if installed).

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull the MMDR 1 (F3) (Left Bus) and/or MMDR 2 (F4) (Right Bus) circuit breaker on the avionics circuit breaker panel and fit safety clips.

c) Using an Allen wrench, turn the securing rod counterclockwise until the unit disengages from the mounting tray.

d) Remove the MMDR unit from the mounting rack.

e) Remove unit from aircraft.

Note Stow the connectors of the removed components.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure that the MMDR 1 (F3) (Left Bus) and/or MMDR 2 (F4) (Right Bus) circuit breaker on the avionics circuit breaker panel are pulled and fitted with safety clips.

c) Using an Allen wrench, turn the securing rod clockwise until it has drawn the unit into the rack and is tight.

d) Remove safety clips and close MMDR 1 (F3) (Left Bus) and/or MMDR 2 (F4) (Right Bus) circuit breaker on the avionics circuit breaker panel.

e) Test the MMDR as per Para 3.A.

5. Troubleshooting

A. Multi-Mode Digital Radio (KTR 2280)

1) If the Multi-Mode Digital Radio (KTR 2280) is deemed inoperative check the following:

a) Ensure the MMDR 1 (F3) and MMDR (F4) circuit breakers are closed.

b) Visually inspect the MMDR units for damaged wires or contacts.

c) Ensure that the MMDR connectors are properly mated and secure.

d) Ensure antenna connections are properly mated and secure.

e) If troubleshooting at the wire level is required refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

f) If repair or service is required, send to appropriate rated avionics repair facility.

Note The MMDR (KTR 2280) contains BITE (Built-In Test Equipment) to enable the operational health of the unit to be constantly monitored. When a critical fault is detected, the KTR 2280 will notify the Apex maintenance system, which will record the fault. The KTR 2280 also stores detected failures in non volatile memory for later review.

Note If appropriately rated for avionics repair, shop fault isolation procedures are contained in the KTR 2280 CMM.

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